EXPERIMENT 8

Configuring Role-Based Access Control (RBAC) in Zoho Creator

Design and Implementation of Permissions, Roles, Role Hierarchy and Data Sharing for the EcoCycle Application

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Aim

To configure role-based access control for the EcoCycle – E-Waste Collection & Recycling Management application through permission sets, a three-level role hierarchy and a Read Only data-sharing rule, so that contact details and records are seen only by authorised staff.

Objectives

- To study the User Permissions section (Permissions, Roles and Data Sharing tabs) of the EcoCycle application.
- To define the permission sets Admin_Access, Read and Write.
- To define the role hierarchy E-Waste Manager → Collection Coordinator → Data Entry Executive.
- To configure a data-sharing rule from Collection Coordinator to Data Entry Executive with Read Only access on the E-Waste Entry form.
- To understand how RBAC separates data ownership from data visibility.

Software / Tools Required

- Zoho Creator (web-based low-code platform)
- EcoCycle application with the E-Waste Entry, Collection Request and Recycling Tracking forms
- Web browser with an internet connection
- Zoho account with access to the application in Edit mode

Introduction / Theory

Why RBAC for EcoCycle

At present EcoCycle runs on the default access of the owner account; no roles or permission profiles are defined. Every user who can open the application therefore sees all records, including the e-mail addresses and phone numbers of requesters. RBAC assigns rights to roles instead of individual users, supports the principle of least privilege and keeps the access policy easy to audit.

Concept	Description
Users	People added to the application who work with its forms, reports and data.
Permissions	Named permission sets (profiles) defining what a user may do with forms, reports and pages.
Roles	Positions in the hierarchy that decide whose records a user can see.
Role hierarchy	A tree in which a higher role can view the records of the roles beneath it; the reverse does not apply.

Concept	Description
Data sharing	Share rules that extend visibility beyond the hierarchy, with Read Only or Read/Write access.

Application Specifications

Parameter	Detail
Application name	EcoCycle (link name ecocycle)
Forms	E-Waste Entry, Collection Request, Recycling Tracking
Permission sets	Admin_Access, Read, Write
Role hierarchy	E-Waste Manager → Collection Coordinator → Data Entry Executive
Custom sharing rule	Collection Coordinator → Data Entry Executive (Read Only) on E-Waste Entry

Procedure

1. Sign in to Zoho Creator and open the EcoCycle application card on the home page (Figure 8.1).
2. Open the application to view the EcoCycle Dashboard, then select **Edit this application** at the top right (Figure 8.2).
3. Select the **Settings** tab from the top navigation (Design, Workflow, Settings).
4. Under the PERMISSIONS group, select **User Permissions**. The Users page opens with the tabs Permissions, Roles and Data Sharing.

[Insert screenshot: Zoho Creator home page showing the EcoCycle application card]

Figure 8.1: Zoho Creator Home – EcoCycle application

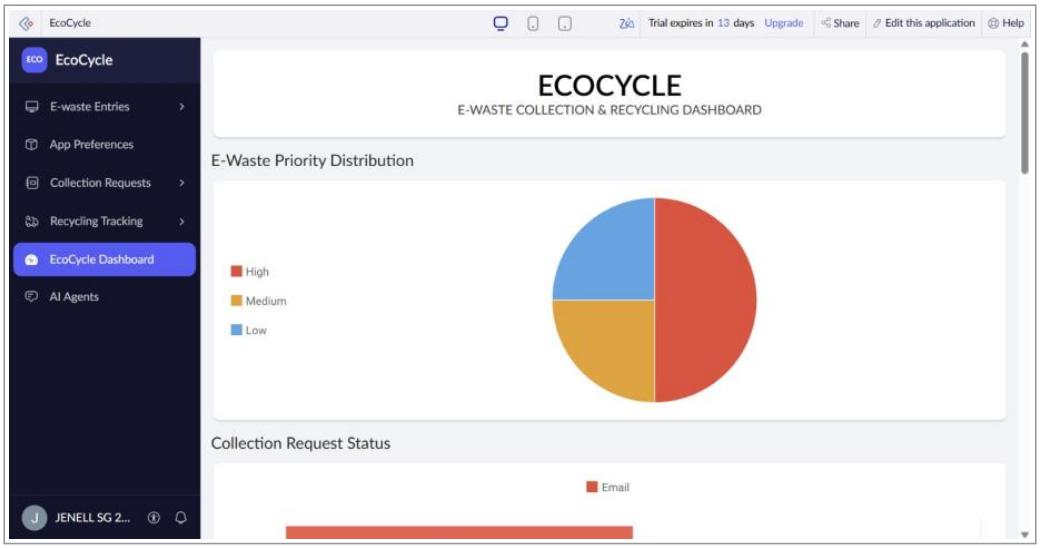


Figure 8.2: EcoCycle Dashboard in access mode (Edit this application at the top right)

1. Open the **Permissions** tab and add the permission sets **Admin_Access**, **Read** and **Write** with the Add Permission button (Figure 8.3).

2. Open the **Roles** tab and arrange the tree E-Waste Manager → Collection Coordinator → Data Entry Executive using Add Role and Manage Role Hierarchy (Figure 8.4).
3. Open the **Data Sharing** tab and choose **New Share Rule**. Enter Name: E_Waste_Entry_Share; Form: E-Waste Entry; Records shared from: Roles (Figure 8.5).
4. Set Records shared from → Roles → Collection Coordinator; Records shared to → Roles → Data Entry Executive; Level of access → Read Only.
5. Select **Save**. The rule appears in the Data Sharing list and can be reopened with Edit Sharing Rule (Figure 8.6).

[Insert screenshot: Settings → User Permissions → Permissions tab]

Figure 8.3: User Permissions – Permissions section

[Insert screenshot: Roles tab showing the three-level role tree]

Figure 8.4: Roles and Role Hierarchy

[Insert screenshot: New Share Rule dialog]

Figure 8.5: New Share Rule – rule configuration

[Insert screenshot: Configured rule in the Data Sharing list]

Figure 8.6: Configured data-sharing rule (Collection Coordinator → Data Entry Executive, Read Only)

Data Sharing

The application follows a layered model: Forms (E-Waste Entry, Collection Request, Recycling Tracking) → Reports → EcoCycle Dashboard → Security layer (Permissions, Roles, Data Sharing). The hierarchy gives upward visibility only: higher roles see the records of the roles beneath them, but

not the reverse.

The custom rule shares E-Waste Entry records from the Collection Coordinator role to the Data Entry Executive role at Read Only level. Data Entry Executives can view the item details they need (name, category, quantity, hazardous flag, condition, location and priority level) but cannot edit or delete them, so the records held by the Collection Coordinator stay the single source of truth. The rule is directional, so access never flows upward or sideways, and it is recorded as one named rule that is easy to audit.

RBAC Working

Level	Role	Intended function
1 (Top)	E-Waste Manager	Overall oversight; sees the records of the roles below; manages roles, reports and the dashboard.
2	Collection Coordinator	Handles collection requests and recycling updates; owns the master e-waste and request records.
3	Data Entry Executive	Records e-waste entries; views shared records read-only.

- Higher roles can view the records of the roles that report to them.
- Lower roles do not gain access to higher-role records unless the Read Only share rule grants it.
- The permission sets Admin_Access, Read and Write further limit what each user may do.

Result

- The User Permissions section of EcoCycle was studied.
- The permission sets Admin_Access, Read and Write were defined.
- The role hierarchy E-Waste Manager → Collection Coordinator → Data Entry Executive was created.
- A Read Only data-sharing rule from Collection Coordinator to Data Entry Executive was configured and saved.

Conclusion

The experiment demonstrates how permissions, roles, the role hierarchy and data sharing together form a role-based access mechanism in Zoho Creator. Applying it to EcoCycle closes the gap noted in the project report: contact details of requesters are no longer visible to every user, and master records cannot be altered downstream. The result is a more secure, auditable and scalable e-waste management application.